Scalable Geometry Optimization for Full-Scale Additive Manufacturing via Reinforcement Learning

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***Abstract*—Recent advances in full-scale additive manufacturing enable high resolution control of structural geometries at previously impossible scales. Such geometric freedom significantly enlarges the feasible design space, allowing structural performance and material efficiency to be optimized beyond traditional approaches. However, this expanded design space introduces substantial computational challenges. Finite Element Analysis (FEA)-based optimization requires repeated large-scale PDE solves, and the cost of iterative evaluation grows rapidly with model resolution and structural scale. For large construction components, conventional multi-iteration topology optimization becomes computationally prohibitive.**

This project proposes a scalable physics-in-the-loop geometry optimization framework that integrates reinforcement learning with parallel FEA solvers built on PETSc. The reinforcement learning agent explores voxel-level geometric modifications, while each design update is evaluated through distributed linear elasticity solvers.

The primary focus of this work is to analyze and optimize the scalability of repeated FEA evaluations within structural optimization iterations and study how this benefits the efficiency and speed if optimization. Performance comparisons across solver configurations and parallel settings will be conducted to quantify the trade-offs between accuracy, efficiency, and optimization progress.

***Index Terms*—Additive Manufacturing, Voxel-Based Optimization, Finite Element Analysis, Reinforcement Learning**

I. Introduction

Full-scale additive manufacturing has recently advanced to the point where voxel-level control of material deposition is technically feasible for architectural and structural components. Unlike conventional construction methods, large-scale 3D printing enables fine-grained customization of geometry, making it possible to locally adjust material distribution to improve stiffness, reduce weight, or enhance structural efficiency. This dramatically expands the geometric design space, moving beyond low-dimensional parametric forms, primarily due to manufacturing constraints, toward high-resolution volumetric representations.

With this increased geometric freedom comes a corresponding increase in computational complexity. Performance-driven structural optimization typically relies on Finite Element Analysis (FEA) to evaluate compliance, stress distribution, or stability constraints. In voxel-based formulations, each design iteration requires assembling and solving large sparse linear systems arising from discretized elasticity equations. For large-scale components, these repeated PDE solves dominate the total computational cost. Traditional gradient-based topology or structural optimization methods, while effective, require numerous iterations and become prohibitively expensive at high resolutions.

The central challenge, which comes before geometric exploration, is the scalability of evaluation.

When optimization requires thousands of structural evaluations, the performance and scalability of the underlying solver become the bottlenecks. Improving the throughput of physics-based evaluation directly impacts the feasibility of voxel-resolution optimization for full-scale structures.

To address this challenge, this project proposes a physics-in-the-loop optimization framework that couples reinforcement learning-based geometric exploration with distributed FEA solvers implemented in PETSc.

The reinforcement learning agent performs localized geometry edits, the structural performance impact of which is evaluated through parallel linear elasticity solves. The primary objective of this work is to study the scalability and efficiency of repeated PDE evaluations within an optimization loop.

Specifically, we aim to analyze solver configuration choices and scaling behavior across multiple MPI processes and prototypes.

By measuring reward-evaluation throughput and parallel efficiency, we investigate how solver design influences optimization progress for voxel-based structural design. This work seeks to bridge large-scale additive manufacturing structure design, physics-based structural evaluation, and high-performance computing, providing a scalable framework for performance-driven geometric optimization.

II. Planned Methodology

We plan to represent large-scale printable components on a voxel grid (or a 3D image with multiple channels) and optimize material distribution under prescribed loads and boundary conditions.

To handle the high-dimensional design space, we will explore learning-based update rules, with a 3D U-Net (CNN variant) as a primary candidate to predict spatial geometry modifications (e.g., add/remove tendencies).

The policy network is parameterized by a 3D U-Net architecture. The network utilizes an encoder-decoder structure with skip connections to capture both the global structural context (e.g., the distance between a load and a support) and local voxel-level details (e.g., surface roughness). The output is a continuous action tensor, A_t, of the exact same dimensions as the input domain, with values bounded within $[-1, 1]$. The structural geometry is then iteratively updated via:

$$\rho_{t+1} = \text{clip}(\rho_t + \alpha \cdot A_t, 0, 1)$$

where ρ represents the voxel density field and α is the step size for the geometric update.

Structural performance will be evaluated through repeated FEA solves implemented with PETSc plus some heuristic constraints, and the project will focus on improving and analyzing the scalability and efficiency of these physics evaluations within the optimization loop.